The Quantized Hall Effect

This variation on a classical phenomenon makes it possible, even in an irregular sample, to measure fundamental constants with an accuracy rivaling that of the most precise measurements yet made

by Bertrand I. Halperin

Few areas of modern science have been studied as intensively as the physics of semiconducting materials. Semiconductors are key components of the electronic devices that are the hallmark of our era. In developing these devices, investigators have learned how to grow nearly perfect semiconductor crystals: how to modify the electronic properties of such crystals by the addition of a few parts per million of foreign atoms, and how to understand so well the properties of the modified crystals that they can incorporate them into circuit elements whose thickness is measured in tens of atomic layers and whose area is so small that a million individual elements can fit on a chip measuring one square centimeter.

It might seem unlikely, then, that in the 1980's semiconductor physicists should have been caught by surprise by the experimental discovery of a dramatic new effect, so little anticipated that many months would pass before experts in the field could develop a satisfactory explanation. In fact, there have been two such surprises.

In 1980 Klaus vo" Klitzing, working at the High Magnetic Field Laboratory of the Max Planck Institute in Grenoble, made a discovery that was to ear" him the 1985 Nobel prize in physics: he observed a phenomenon that is now called the integral quantized Hall effect. Two years later, in an experiment at the National Mag-"et Laboratory in Cambridge, Mass,, Daniel C. Tsui, Horst L. Störmer and Arthur C. Gossard of AT&T Bell Laboratories observed a second remarkable phenomenon, now known as the fractional quantized Hall effect. As their names imply, these effects have much in common, but they require significantly different explanations.

The quantized Hall effect, integral or fractional, is actually a special case of a more general phenomenon known as the Hall effect. To observe the Hall effect, a planar sample of conductive material is placed in a magnetic field perpendicular to the sample's surface. Then, by means of electrodes at each end of the sample, a small electric current is made to pass from one end to the other. (An electric current is simply the motion of charged particles; in the case of most solids the current-carrying particles are electrons.)

Two measurements are then made. To make one measurement, contacts "ear both ends of the sample are connected to a voltmeter, which measures the voltage drop from one end of the sample to the other, that is, in the directio" parallel to the Row of current. (This is essentially a measure of the difference between the energy of the electrons entering the sample and the energy of those that have passed through it.) The other measurement is made by a voltmeter attached to electrodes positioned on opposite edges of the sample in such a way that a line drawn between the electrodes would be perpendicular to the direction of the current. This voltmeter measures the voltage difference across the sample, in the direction perpendicular to the current.

If there were no magnetic field, the voltage difference perpendicular to the current would be equal to zero. As the American physicist Edwin H. Hall first noticed in 1879, when there is a magnetic field perpendicular to the sample, a voltage difference occurs that is in most cases directly proportional to the strength of the magnetic field. The voltage drop parallel to the current is usually about the same whether or not there is a magnetic'tield.

The voltage drop parallel to the current and the voltage drop perpendicular to the current are both directly proportional to the amount of current flowing in the sample. It is therefore convenient to divide each of these voltages by the amount of current flowing and thus obtain quantities that arc independent of the current flow. The ratio of the voltage drop along the sample to the current is just the ordinary electrical resistance of the sample. The ratio of the voltage drop across the sample to the current flowing along it is called the Hall resistance.

In his 1980 experiment von Klitzing was studying the Hall effect in certain devices in which the electrons free to carry current are confined within a thin layer. He found that when such devices are cooled to within about a degree of absolute zero and placed in extremely strong magnetic fields, the behavior of the ordinary resistance and the Hall resistance was dramatically different from the behavior described by Hall. He had found the integral quantized Hall effect.

The most striking difference between the integral quantized Hall effect and the normal Hall effect is that the Hall resistance, instead of increasing steadily and linearly as the strength of the magnetic field is increased, exhibits a series of plateaus. That is, there are intervals in which the Hall resistance appears not to vary at all when the strength of the magnetic field is varied. (Between such plateaus, the Hall resistance does rise smoothly with increasing magnetic field.) In addition, in the same intervals of magnetic field strength where the Hall resistance exhibits plateaus, the voltage drop parallel to the current seems to disappear completely. That is, there is no electrical resistance in the sample and current flows from one end to the other without dissipating any energy.

The vanishing electrical resistance and the plateaus in the Hall resistance are remarkable phenomena in themselves. The effect seems eve" more astonishing when one examines the val-"es at which the plateaus in the Hall resistance appear. On each plateau the value of the Hall resistance satisfies a remarkably simple condition: the re-

ciprocal of the Hall resistance is equal to an integer multiplied by the square Of the charge on the electron and divided by Planck's constant, the fundamental constant of quantum mechanics. Each plateau is characterized by a different integer.

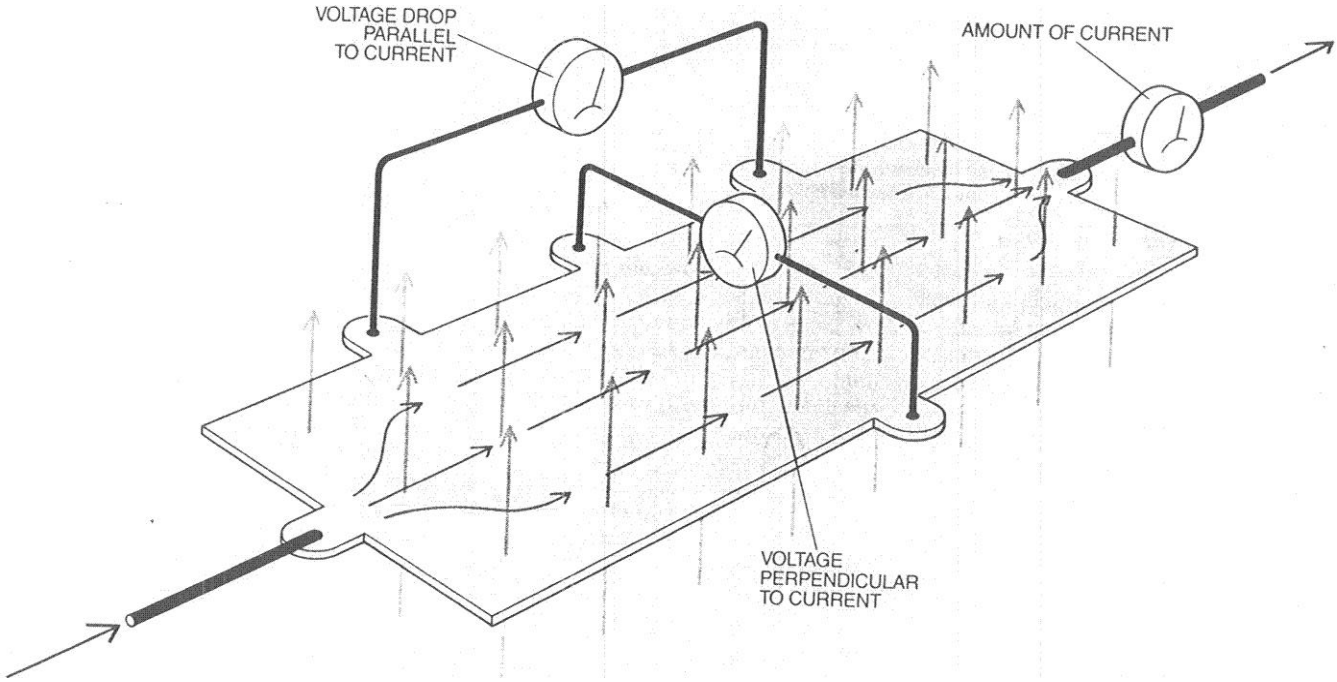
The fractional quantized Hall effect found by Tsui, Störmer and Gossard occurs only in devices that are exceptionally free from defects that interfere with the motion of electrons. In the fractional effect there are additional plateaus where the reciprocal of the Hall resistance is equal to a simple fraction (rather than an integer) multiplied by the square of the charge on the electron and divided by Planck's constant. In the original experiments the observed fractions were 1/3 and 2/3. More recent experiments have found plateaus at levels that correspond to such other fractions as 4/3, 5/3, 2/5 3/5 and 3/7, but no plateaus have been found at levels corresponding to fractions with even denominators.

What is, most amazing about the quantized Hall effect is its degree of precision. Measured values of the Hall resistance at various integer plateaus satisfy the simple relation I have described with an accuracy of about one part in six million-precisely the accuracy with which the values of the fundamental constants themselves are known. That the Hall resistance in the integral effect is precisely quantized (in other words, that individual values of the Hall resistance are equal to the same quantity multiplied by different integers) has been examined by determining the ratio of the Hall resistances at different plateaus and ascertaining that they are in the ratio of integers. The experiments that have determined such ratios are accurate to within one part in 30 million.

The precision of the quantized Hall resistance is particularly surprising because under normal conditions the Hall resistance of a sample, like the ordinary electrical resistance is notoriously sensitive to such details as the sample's geometry, its processing history, the amount of impurity it contains and its temperature at the time of the experiment. The quantized Hall effect, on the other hand, is the same for samples with different geometries and processing histories, and even for those made of a variety of materials. It is almost without precedent for any property of a solid to be reproducible from one sample to another at the levels of accuracy attained by the quantized Hall effect.

The integral and fractional quantized Hall effects are found only in semiconductor devices in which the electrons that are free to carry current are confined within a thin layer of the semiconductor. There are two common devices that constrain electrons in this way. In the original device studied by von Klitzing, which is called a field-effect transistor, a metal contact is separated from a silicon substrate by a thin layer of insulating material. A voltage applied to the metal contact can attract electrons to a thin layer just under the surface of the silicon. (The high-quality samples employed by von Klitzing were constructed by Gerhardt Dorda of the Siemens Research Laboratory in Munich and Michael Pepper of the University of Cambridge.)

The device in which the fractional quantized Hall effect was first seen, which is called a heterojunction, is an interface of crystals made out of two different semiconducting materials. In a heterojunction, electrons from



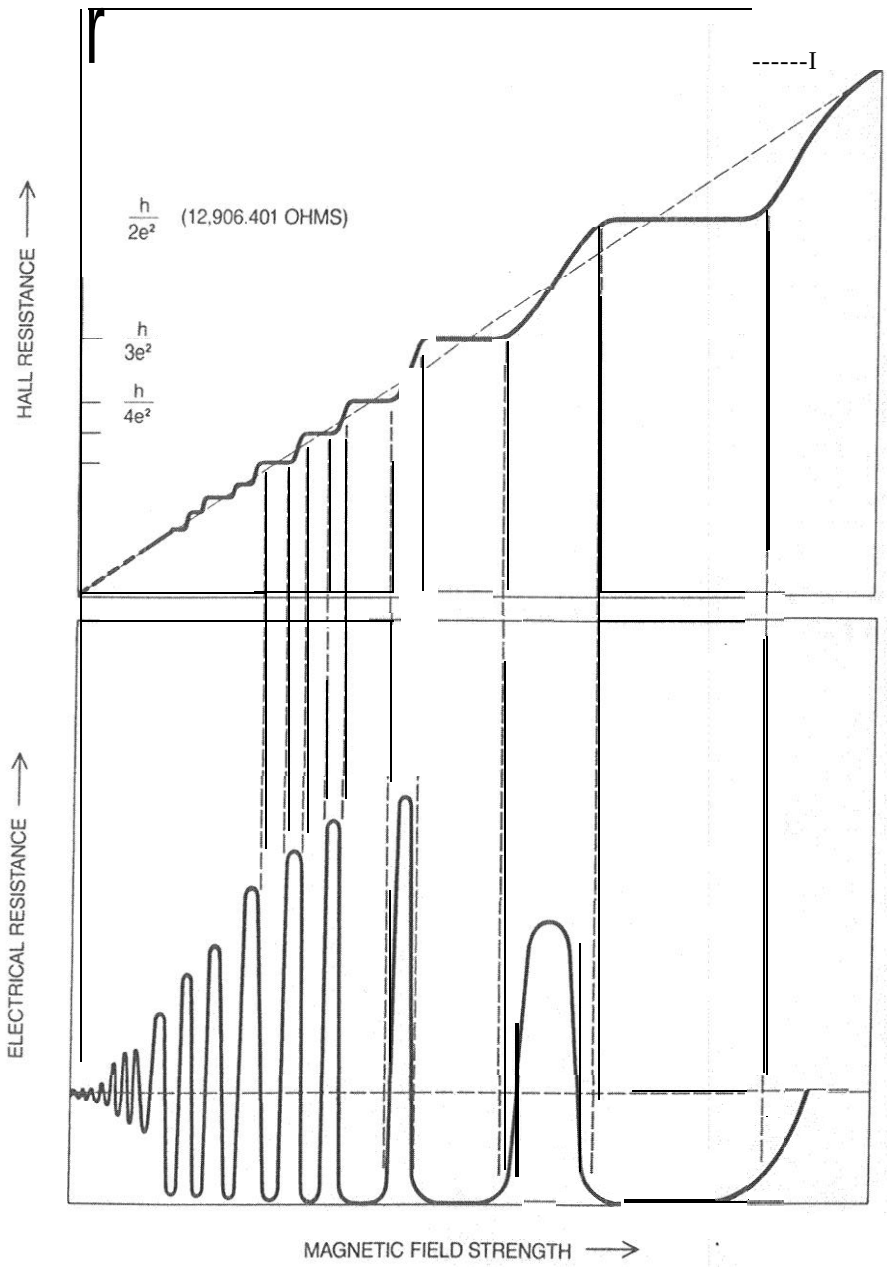
APPARATUS for measuring the Hall effect consists of a flat plane of conductive material and devices for measuring voltages and currents. A measured amount of current *(horizontal arrows)* is made to flow from one end of 'he sample to the other and a magnetic field *(vertical arrows)* is applied perpendicular to the sample. Two voltages are measured: the voltage drop from one end of the sample to 'he other (which is essentially the difference in energy between the electrons entering the sheet and those leaving it) and the voltage perpendicular to the current which is called the Hat, voltage. In the classical Hall effect the Hall resistance (the Hall voltage divided by the amount of current Rowing) is directly proportional to the strength of the magnetic field: it increases steadily as the magnetic field strength is increased. In the quantized Hall effect, which is seen when the electrons that carry current are confined to a thin layer of a semiconductor device at low temperatures and in extreme. ly strong magnetic fields, the Hall resistance shows plateaus: there are intervals of magnetic field strength in which the Hall resistance remains constant as the magnetic field is varied. Moreover, in those intervals 'he sample's electrical resistance (the voltage drop parallel to the current divided by the amount of current) disappears completely, indicating that current flows through the sample without dissipating any energy The values of the Hall resistance on the different plateaus are exactly the same regardless (within broad limits) of the geometry or even the composition of the device.

one semiconductor are attracted to more energetically favorable locations in the other semiconductor. The positive charge thereby created in the "donor" semiconductor provides a force attracting the electrons back, however, and they become trapped in a thin layer at the interface of the two crystals.

In both the field-effect transistor and the heterojunction the current-carrying electrons are free to move anywhere within the plane to which they are confined. Motion perpendicular to this plane, however, is almost completely impossible, and so for all practical purposes the layer Of electrons may be regarded as two-dimensional. In experiments on the quantized Hall effect this thin layer of electrons plays the role the flat sheet of conductor plays in normal Hall experiments.

Although the properties of two-dimensional layers of electrons have been studied thoroughly, the integral quantized Hall effect was a great surprise when it was discovered. Some aspects of the effect had actually bee" predicted as a" incidental result in a paper on a related subject written in 1975 by Tsuneya Ando, Yukio Matsumoto and Yasutada Uemura of the University of Tokyo, but the most important aspect, the extreme precision of the effect, had not been expected because the predictions were based on a type of analysis that is in general only approximately correct for the electrons in real materials. For the fractional quantized Hall effect there was not even an approximate explanation available at the time of its discovery.

Although there are now several convincing explanations Of the exactness of the quantized Hall effect. it is not possible to present in this article a complete explanation Of the fractional Or the integral quantized Hall effect. Any such explanation would necessarily be written in the highly mathematical language of quantum mechanics, which is the basis for all modern descriptions of phenomena on the atomic scale. It is possible, however, to give a partial explanation that incorporates many of the essential ideas and that will at least make plausible the experimental results.



QUANTIZED HALL EFFECT appears as plateaus in the Hall resistance of a sample *(top)*, which coincide with the disappearance of the sample's electrical resistance *(bottom)* On the plateaus the Hall resistance does not change when the magnetic field strength is varied. In contrast, in classical experiments (color) the Hall resistance increases linearly with increasing magnetic field strength and the resistance is constant. At each plateau the value of the Hall resistance is precisely equal to Planck's constant (the fundamental constant of quantum mechanics, designated by the letter h divided by an integer multiple of the square of the charge on the electron. The quantized Hall effect thus makes possible precise measurements of fundamental constants and extremely accurate calibration of instruments

Essential to any exposition of the Hall effect are such basic classical concepts as electric and magnetic fields. An electric field is a field that exerts a force on charged objects, much as a gravitational field exerts a force on massive objects. When a marble rolls down a hillside, it tends to roll in whatever direction the hill is steepest; similarly, a charged particle in a" electric field generally feels a force in the direction of the field lines.

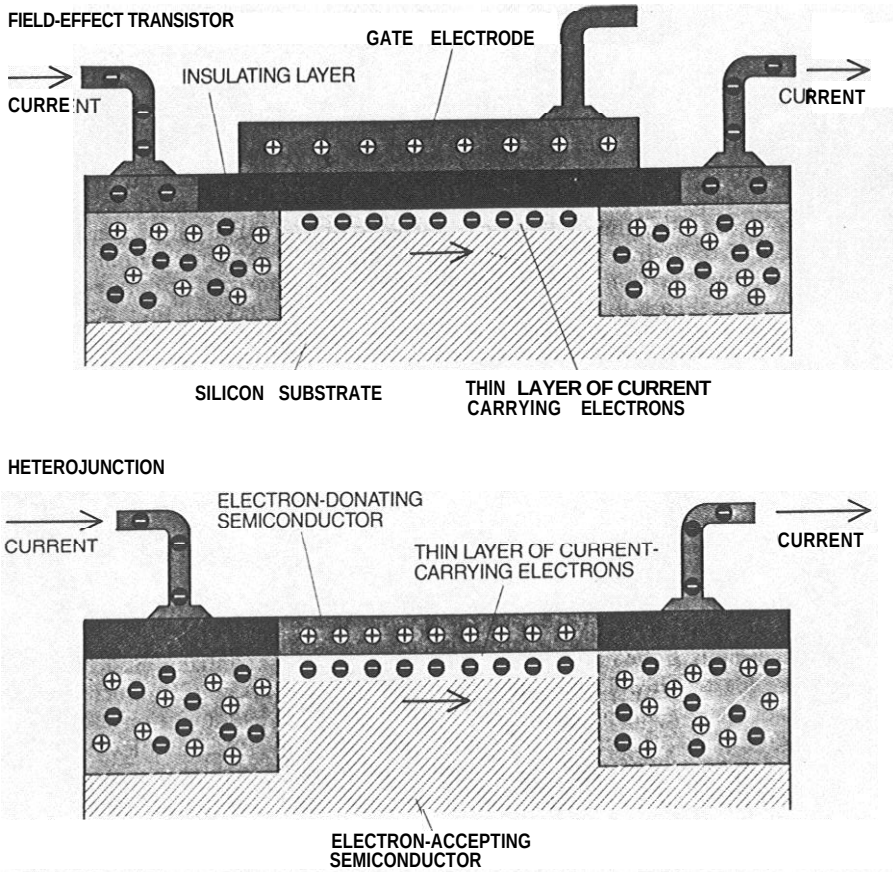
Just as a gravitational field may be created by a concentration of massive objects, so a" electric field may be created by a concentration of charged particles. The analogy can be carried further: just as an object high On a hill has a greater "gravitational potential energy" than a" object in a valley, so a charged particle may have a greater "electrical potential energy" than a similar particle elsewhere in a" electric field. The difference between the electrical potential energy of similar charged particles at two different points in space is called the voltage difference between those points.

Unlike an electric field, a magnetic field exerts a force **only on moving** charged particles. The magnitude of the force exerted on a particle directly proportional to the particle's velocity and the strength of the magnetic field. The direction of the force is per-

pendicular to both the particle's direction of motion and the direction of the magnetic field. A moving charged particle in a magnetic field therefore moves in a circle: at each instant the magnetic field exerts a force perpendicular to the particle's motion, and so at each instant the particle's path curves. The radius of the circular orbit is inversely proportional to the strength of the magnetic field, so that stronger magnetic fields hold particles in smaller orbits.

Suppose a moving charged particle is placed simultaneously in both an electric field and a magnetic field. In this case the particle follows a cycloidal path: it moves in a circle around a so-called guiding center, which itself moves in a. straight line ***[see illustration on next page]*** It turns out that the guiding center of the particle's motion moves in a line perpendicular to both the electric field and the magnetic field. In other words, it moves neither up nor down the "hillside" of the electrical potential energy. Rather, on the average it tends to move horizontally along the hillside, in the direction perpendicular to that of the slope.

The speed with which the guiding center moves, called the drift velocity, is inversely proportional to the strength of the magnetic field-when the magnetic field is stronger, the particle's motion is more circular and less linear-and directly proportional to the strength of the electric field; thus the guiding center moves faster in regions where the "incline" is steepest, even though it moves along the slope, not down it.



SEMICONDUCTOR DEVICES in which the quantized Hall effect is observed hold current-carrying electrons within a thin layer of semiconducting crystal. In a field-effect transistor (*top*) electrons are attracted to the surface of a silicon crystal by a positive charge applied to a metal contact called the gate electrode. Insulating material prevents the electrons from entering the electrode, and they remain in a thin layer near the crystal's surface. In a heterojunction (*bottom*), an interface of semiconductors with differing electronic properties, electrons from one semiconductor are attracted to lower-energy states in the other. The positive ions left behind attract the electrons back, however, trapping them in a thin layer.

With these basic principles it is possible to understand the classical Hall effect. Suppose a current of electrons is flowing from left to right on a finite sheet of metal that lies on a plane parallel to this page. Now suppose a magnetic field is applied perpendicular to the metal sheet. At first the electrons will be pushed in a direction perpendicular both to their original direction and to that of the magnetic field: they will move toward the top or bottom edge of the page. If electrons cannot leave the metal except through the contacts at the left and right ends, they will accumulate at one edge (top or bottom) of the sheet. This accumulation of charged particles produces an electric field that points from one edge of the metal to the other, and so one edge will be at a higher voltage than the other.

The electric field, in combination with the magnetic field, impels electrons to move in cycloidal paths in a direction perpendicular to both fields: from left to right. The voltage difference between the two edges of the metal is therefore perpendicular to the direction in which current flows.

If the strength of the magnetic field is increased, then for a given amount of current to flow (that is, for electrons to continue to flow from left to right at the same rate) the strength of the electric field-and the voltage difference between the metal's edges-must also increase. The reason is that the "drift velocity" with which the guiding center of each electron's cycloidal path moves from left to right is inversely proportional to the strength of the magnetic field and directly proportional to the strength of the electric field. Consequently when the strength of the magnetic field is increased, more electrons are deflected toward one edge of the metal, creating a stronger electric field and a greater voltage difference between the edges. The voltage difference, which increases linearly with an increase in the strength of the magnetic field, is the classical Hall effect.

This kind of classical analysis will not explain the quantized Hall effect, which is inherently a quantum-mechanical phenomenon. It is possible to understand some aspects of the quantized Hall effect, however, by means of what is called semiclassical analysis, a mixture of classical and quantum-mechanical ideas that avoids the mathematical complexity of a full quantum-mechanical description. Although a semiclassical analysis is not fully reliable, it is nonetheless effective as a guide that suggests the general form of the quantum-mechanical answer, and in fact it is often relied on for this purpose by experts in the field.

In the classical case an electron in a uniform magnetic field follows a closed circular orbit. In quantum mechanics the energy of a particle in such a closed orbit can have only certain discrete values, just as an electron in an atom can occupy only certain energy levels. The energy levels of the two-dimensional motion of a charged particle such as that of an electron in a uniform magnetic field are known as Landau levels. There is no discrete set Of

energy levels for thick, "three-dimensional" systems in a uniform magnetic field, because the energy of the particle's motion parallel to the magnetic field, unlike that of its motion in the plane perpendicular to the field, is not quantized. It is therefore significant, for the sake of this analysis, that the actual physical system under consideration consists of a thin, "two-dimensional" layer of electrons.

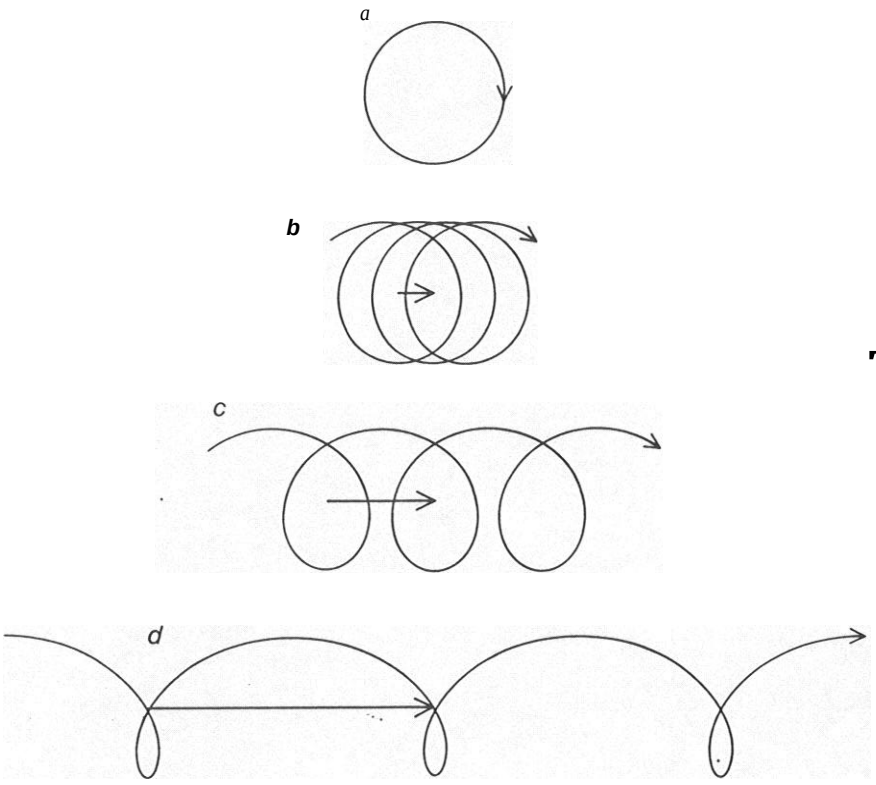
In the classical case there are many places within the plane where the electrons' identical circular orbits could be centered without overlapping one another. Likewise, in the semiclassical case there are a number of states with the same energy that are independent of one another. (For each state there is a distinct "wave function," a mathematical description of the electron's probable position.) Specifically, the number of independent states per unit of area is directly proportional to the strength of the magnetic field. This is at least plausible, because in the classical case a stronger magnetic field confines the electrons to tighter orbits, and so more orbits can fit in a given area without overlapping. It turns out that the constant of proportionality between the number of independent states per unit of area and the strength of the magnetic field depends only on Planck's constant and the charge of the electron.

So far I have ignored an aspect of the actual experimental system that is very important in all properties involving electrical conduction: the influence of impurities or defects in the semiconductor crystal that holds the thin plane of conductionelectrons. Impurities play a particularly important role in the ordinary electrical resistance of metals and semiconductors. Much of the energy dissipation that characterizes resistance occurs when electrons are scattered by collisions with impurity atoms or defects in the crystal lattice. Paradoxically, the presence of impurities is what leads to the disappearance of electrical resistance and the plateaus in Hall resistance that constitute the quantized Hall effect.

In the presence of impurities the many independent quantum states that make up a given Landau level are no longer precisely equal in energy. In a semiclassical explanation one might say, for example, that in certain of the quantum states the electron is slightly more likely to be found near an impurity atom that has, say, an excess of positive charge. (This would be the case if the impurity atom were of an element that tends to give up an electron when it is buried in a semiconductor lattice.) Such states would be slightly stabler than others in the same Landau level and would have a slightly Iower energy. The single energy level that makes up a. Landau level in a pure crystal is thus spread out, in the presence of impurities, into a band made up of many distinct energy levels.

The various quantum states in each energy band can be divided into three general classes. The states near the bottom of each band, that is, those of lowest energy, are each localized in some small region of the sample. An electron in such a state would never leave that region. These low-energy localized states occur at "valleys" in the electrical potential energy, for example in the regions around impurity atoms that have an excess of positive charge. Similarly, near the top of the band are high-energy localized states. These are localized near "peaks" in the electric potential, perhaps in the regions around impurity atoms that have acquired electrons and so bear an excess of negative charge.

Near the center of each energy band are the so-called extended states, each of which is spread out over a large region of space. The distinction between localized and extended states is important: because electrons in localized states cannot move very far within the sample, they cannot carry current *[see illustration on pages 58 and 59]*



CYCLOIDS describethe motion of an electron in combined electric and magnetic fields In the absence of an electric field an electron in a magnetic field moves in a circle in the plane perpendicular to the field (a) When an electric field is applied, the electron follows a cycloid: it moves in a circle around a guiding center that itself moves in a straight line. The tine followed by the guiding center is perpendicular to both the electric field and the magnetic field. The speed with which the guiding center travels is inversely proportional to the strength of the magnetic field and directly proportional to the strength of the electric field; when the electric field is stronger or the magnetic field is weaker (b-d), the electron's motion is more linear and the guiding center moves with greater "drift velocity" *(colored arrows).*

Taking into consideration one more aspect of the actual experimental system, the extremely low temperatures at which the experiments are conducted, will now make it possible to give some explanation of the integral quantized Hall effect.

At very low temperatures the electrons in the sample take on the configuration that has the lowest possible energy. According to the Pauli exclusion principle, one of the foundations of quantum mechanics, no two electrons can occupy the same quantum-mechanical state. When the system is at its lowest possible energy, then, each available quantum state below a certain energy level contains exactly one electron and each state above that level contains no electrons. The energy of the highest filled energy level is known as the Fermi energy.

When there is a voltage difference between the two edges of the sample, it is not actually possible to define a sin-

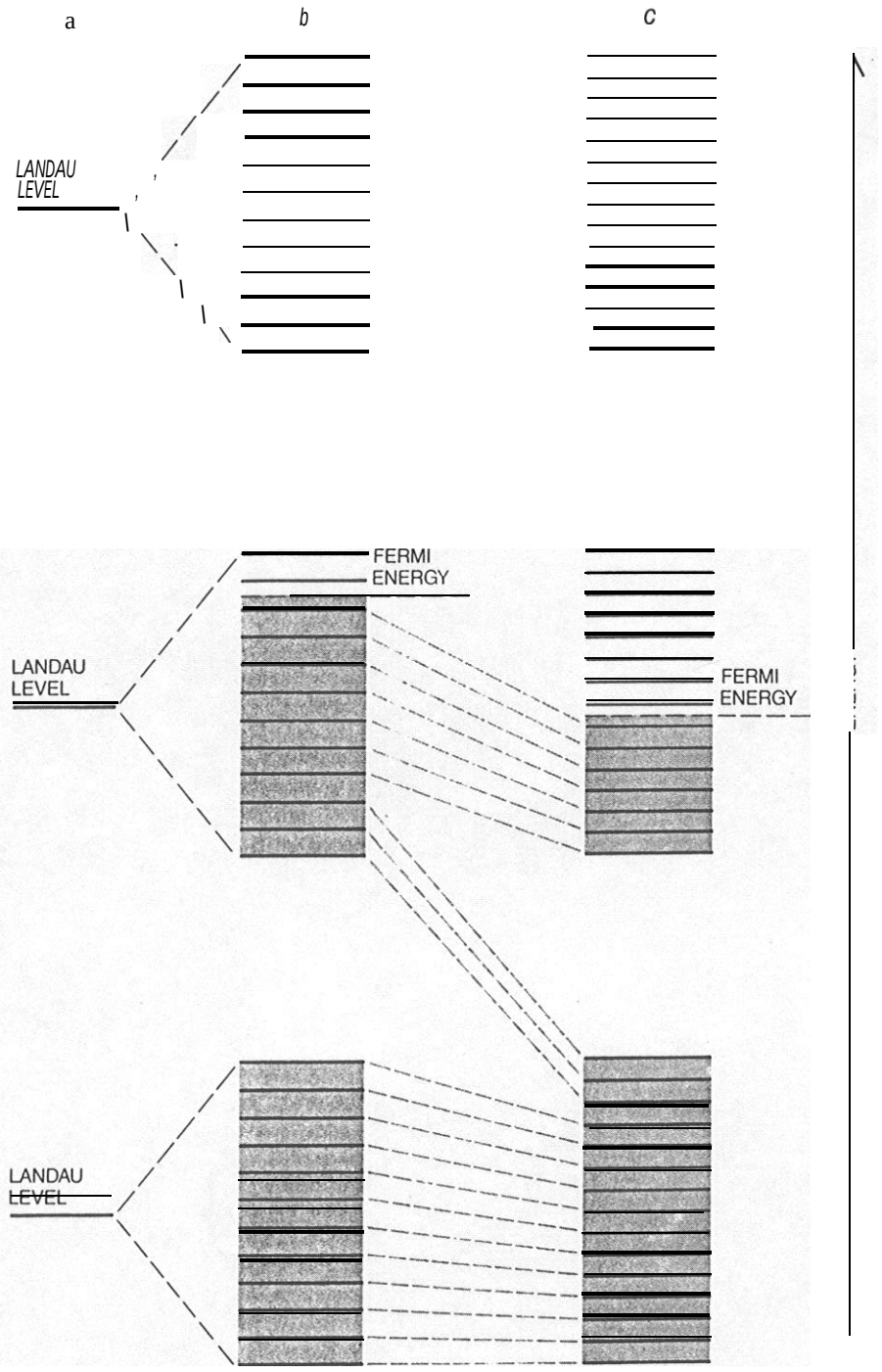
gle Fermi energy for the entire plane of conduction electrons; the Fermi energy varies slightly from point to point. In each region of space the states with energies below the local Fermi level are occupied and states above it are empty. The voltage measured by a voltmeter connected to two points on opposite edges of the sample is actually a measure of the difference between the local Fermi energies of the two points; the difference between the local Fermi energies represents the difference between the energies of the highest-energy electrons at each point.

Suppose a current is flowing along a sample in a perpendicular magnetic field and the Fermi level of the sample's electrons is in the sub-band of localized states near the top of some Landau band. In this case all the extended states and the low-energy localized states of the Landau band will be completely filled, and some of the high-energy localized states will be occupied as well. Now suppose that the strength of the magnetic field is gradually increased and that at the same time the current is continuously adjusted in such a way that the Hall voltage between the sample's two edges remains constant.

Because the number of independent quantum states per unit of area is directly proportional to the applied magnetic field, as the magnetic field increases, the number of independent quantum states in each Landau level increases proportionately: in every region of space within the sample, additional quantum states that have roughly the same energy as neighboring states become available.

Many of the newly available states will be below the local Fermi level, and so electrons from higher-energy occupied states will drop down to fill them; these electrons will in general come from the high-energy localized states that are near the Fermi level. As these states are vacated the Fermi level-the energy of the highest occupied state-descends to a lower position within the Landau band *[see illustration at right]*. As long as the Fermi level remains in the sub-band of high-energy localized states, all the extended States within the Landau band remain fully occupied.

Because an electron trapped in a localized state cannot move through the sample, the changing fraction of localized states that are filled has no effect on the sample's large-scale electrical properties. The amount of current flowing in the sample therefore remains constant as long as the sub-band of extended states is completely filled: although the increased magnetic field slows the forward motion of any cur-



ENERGY LEVELS available to an electron confined in a two-dimensional layer in a perpendicular magnetic field determine the layer's large-scale electrical properties. In a layer in an ideal crystal (a) an electron can exist only in states that have certain energy levels, catted Landau levels. In real crystals, which always contain impurities each Landau level is spread into a band cd energy levels (b). states in the same Landau level, which had once bad equal energies, come to have slightly different energies because in some states the electron is closer to an impurity atom. The energy levels are of two types: localized states (color) in which the electron is bound in the vicinity of an impurity atom or defect, and extended states, in which the electron can roam over a large area of the crystal. Localized states can have high or tow energies, depending on the charge of the impurity producing them. Only electrons in extended states can carry current, and so the crystal's electrical properties are determined by the electrons in extended states. Because of the Pauli exclusion principle, each state may contain no more than one electron. At very tow temperatures every state below a certain energy, the Fermi energy, is occupied *(shaded region)* and every state above it is empty. When the strength of the magnetic field is increased, there are more states in each Landau band, and electrons drop *(broken lines)* into states with tower energies (c). The Fermi energy thereby drops to a tower position within the Landau band or to a tower Landau band.

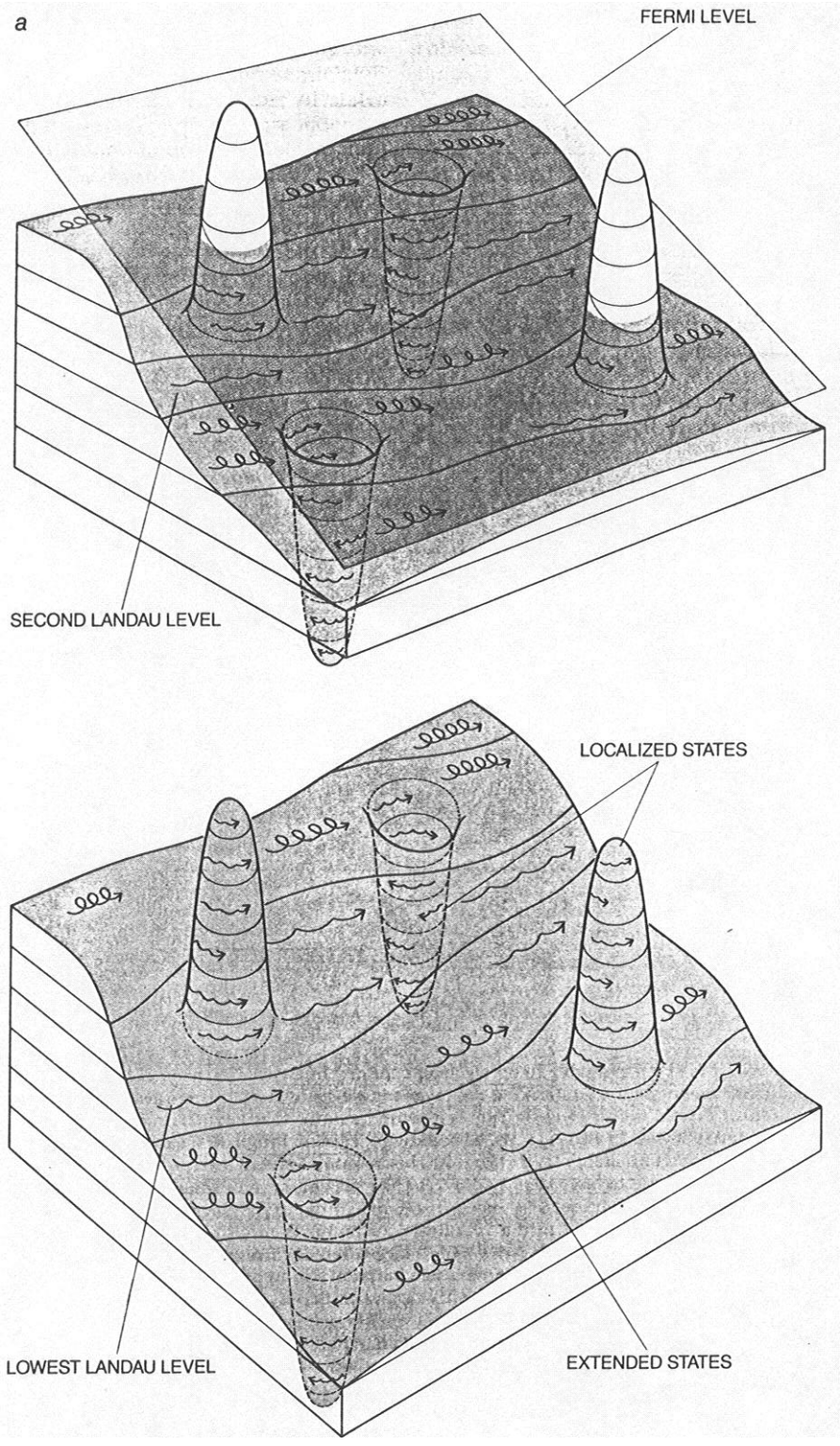
rent-carrying electrons, this effect is precisely canceled by the increase, due to the newly created extended states, in the number of electrons available to carry current.

Since the Hall voltage is being held constant, the fact that the current does not change as the magnetic field is varied implied that the Hall resistance (the Hall voltage divided by the amount Of current flowing) also remains constant. (A" equivalent experiment, in which the current is externally held constant and the Hall voltage is measured, can also be performed; in this experiment the Hall voltage, and hence the Hall resistance, remains constant whenever the Fermi level is in the sub-band of localized states.) Whenever the Fermi level is in the sub-band of localized states, then, the Hall resistance remains the same eve" when the magnetic field is varied. This is the plateau in Hall resistance that is characteristic of the quantized Hall effect.

Eventually, as the strength of the magnetic field is increased, the supply of electrons in the high-energy localized states will be exhausted and the Fermi level must drop into the sub-band of extended states. As the Fermi level descends through the sub-band of extended states, some of them are vacated. Because the current-carrying sub-band is then only partially occupied, the amount of current Rowing decreases and the Hall resistance therefore increases. The Hall resistance continues to increase when the magnetic field is increased, as long as the Fermi level remains in the sub-band of extended states.

If the magnetic field is increased further, eventually the extended states within the Landau band will all be emptied, and the Fermi level will once again enter a sub-band of localized states: the low-energy localized states at the bottom of the Landau band. If there is at least one full Landau band below the Fermi level, the extended states in that band will be able to carry a current and the quantized Hall effect will once more be observed. Because the extended states of one Landau band have been completely emptied, however, the number of sub-bands of occupied states has been reduced by one; the electric current is therefore reduced in precise proportion to the number of occupied sub-bands of extended states, and the Hall resistance is accordingly larger than it was on the previous plateau.

In this model it is easy to understand why the ratio of Hall resistances at any two plateaus should equal a ratio of integers. The reason is that for any given Hail voltage the current is



GEOMETRIC REPRESENTATION of the energy of electrons confined to a two-dimensional layer in a strong magnetic field illustrate. the difference between localized and extended states and explains the quantized Hall effect. The vertical axis represents the energy, within a small section of a sample, of electrons in two Landau hands. Peaks and valleys in the electric potential are due to impurities spaced randomly within the sample. Electrons move in cycloidal paths parallel to contour lines (lines of constant electric potential); hence closed contours around peaks and valleys indicate localized states, in which electrons orbit within a small region of space, and open contours indicate extended states, which may stretch from one end of the sample to the other. At tow temperatures electron states whose energy lies below the local Fermi level (*colored plane)* are filled *(shad-*

directly proportional to the number of occupied sub-bands of extended states, and on each plateau an integral number of such sub-bands is filled.

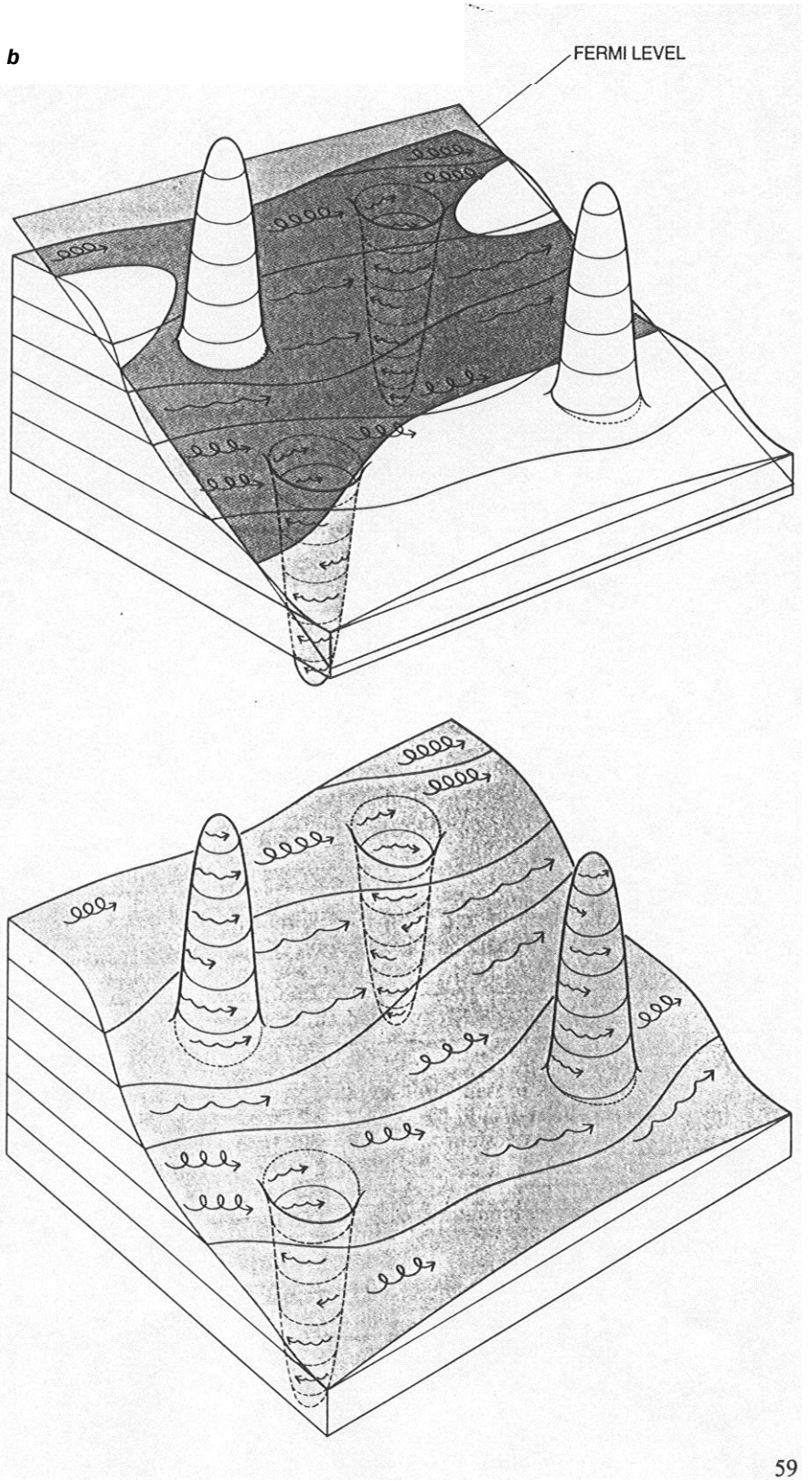
If the magnetic field is increased still further, the Fermi level will move down through the regions of localized states at the bottom of one Landau band and into the high-energy localized states at the top of the next Landau band. The Hall resistance will remain constant at its new plateau value until the Fermi level reaches the region of extended states in the middle of this next Landau band.

The second striking feature of the quantized Hall effect is that current flows through the sample with no resistance, that is, without dissipating any energy. This feature can be explained by the principle of conservation of energy.

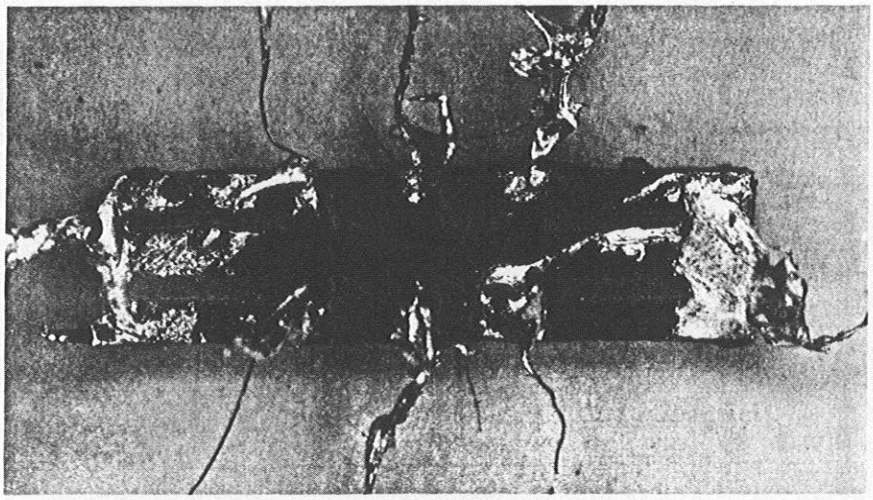
To dissipate power an electron must make a transition from one energy state to a state of lower energy, dissipating the excess energy as vibrational energy, or heat, within the crystal lattice. Because two electrons cannot occupy the same quantum-mechanical state, the new state into which the electron falls must be one that was previously empty. If the Fermi level is in the sub-band of energies corresponding to the extended states, an electron can find itself in an extended state that is below the local Fermi energy in one region of space but that extends into a region where its energy is above the local Fermi energy. In that region there may be states with energies between the energy of the electron's state and the local Fermi energy. The electron would then be able to drop into one of those states.

In particular it could happen that an electron enters the sample in a region where the local Fermi energy is high and exits from the sample in a region where the local Fermi energy is lower. Such an electron might drop into a lower level as it passes through the sample, exiting with less energy than it had on entering, having dissipated some of its energy as heat. The sample would thus exhibit electrical resistance: electrons would lose energy as they passed through it.

If the Fermi Ievel is within the energy levels corresponding to the localized states, however, the sample cannot exhibit electrical resistance. The electrons with the highest energy-those whose energy is near the Fermi level-will all be localized in very small regions of space. In those regions all the unoccupied states must have energies higher than the local Fermi energy. There may be empty states of lower energy in some distant part Of



***ed region*) and others are empty. The surfaces of electron energy and the local Fermi level are tilted from the horizontal because of a voltage difference (the Hall voltage) between the two edges of the sample. When the Fermi level is in a region of localized states (*a*), the quantized Hall effect occurs: the Hall resistance (the Hall voltage divided by the current flow) is not affected by small changes in such parameters as the strength of the magnetic field or the number of electrons in the layer. For example, a small change in the Fermi level would not change the Hall resistance because it would not affect electrons in the extended, current-carrying states. When the Fermi level is in a region of extended states (*b*), on the other hand, a small change in its height does indeed affect the current flow by emptying or filling some of the extended states, and the quantized Hall effect does not occur.**



EXPERIMENTAL DEVICE (a heterojunction) for measuring the quantized Hall effect has irregular features and imprecise geometry, yet it produces results that are accurate to within about one part in six million (the same accuracy with which the fundamental constants of electromagnetism and quantum mechanics are known). The most astonishing aspect of the quantized Hall effect is that, within broad limits, the extreme accuracy of the experimental results is independent of the shape, purity or processing history of the sample.

the sample, but it is impossible for the electron to make a transition between localized states that are far apart in space. The electrons therefore cannot drop into states of lower energy, and so they cannot dissipate energy.

This explanation of the absence of ordinary resistance presupposes one important condition: that there exists at least one Landau band, below the Fermi energy, that has occupied extended states. These states carry the current that passes without dissipation. If the Fermi level descends until all the states below it are localized, no current will be able to flow and the sample will act as a perfect insulator, not a perfect conductor.

The localized states created by the presence of impurity atoms and defects are crucial to this explanation of the quantized Hall effect. The localized states act as a reservoir for electrons, so that, for a range of magnetic field strengths, the extended states in each Landau band are either completely empty or completely filled. Without such a reservoir the resistance would vanish only when there were precisely enough electrons present to fill a Landau level and there would be no plateaus at all in the Hall resistance.

The argument I have given so far represents a plausible explanation of the integral quantized Hall effect, but the fractional quantized Hall effect is much more difficult to explain. The integral effect is seen most readily when an integral number of Landau levels are filled. The fractional effect, on the other hand, is seen only when a Landau level is partially filled. For example, a plateau on which the reciprocal of the Hall resistance is equal to one-third of the square of the electron's charge divided by Planck's constant is seen when the lowest Landau level is approximately one-third full.

Even a completely quantum-mechanical treatment of the model I have adhered to so far-that of independent electrons moving under the influence of electric and magnetic fields-shows no special stability when a fraction of the available states are filled. To explain the fractional quantized Hall effect it is necessary to take into account the mutual interactions of electrons, and this means one must apply a quantum-mechanical wave function that depends simultaneously on the positions of all the particles in the system.

The most important breakthrough in understanding the fractional quantized Hall effect came in 1983, when Robert Laughlin of the Lawrence Livermore National Laboratory constructed a wave function that had the required stability when the fraction of filled states had such values as 1/3, 1/5, 1/7, 2/3, 4/5 or 6/7 Each of these fractions is equal to 1 divided by an odd integer or it is equal to 1 minus such a fraction. The importance of this development is underscored by the fact that the wave functions constructed to explain the fractional quantized Hall effect are radically different from any wave functions previously encountered in physics.

The denominator is. odd in each of the stable fractions in part because of a quantum-mechanical requirement that is related to the Pauli exclusion principle but concerns a component of the wave function called its phase, for which there is no classical analogue.

After Laughlin's original work, other theorists, including Laughlin and Duncan Haldane, who was then at the University of Southern California, and I were able to propose wave functions that are stable for other fractions having odd denominators, such as 2/5, 2/7 and 3/7.

An interesting theoretical prediction resulting from Laughlin's analysis is that if one adds an extra electron to a system in which the Landau level is already one-third full, the extra charge should appear in three separate places in the sample. Precisely one-third of the charge on the electron should appear in each place. Similar fractional charges should appear if one adds or removes electrons at any of the fractional quantized Hall plateaus. The fractional charges are called quasiparticles, and they are expected to behave very much like charged particles.

In the fractional quantized Hall effect it is the states of the fractionally charged quasiparticles, rather than those of electrons, that are localized by the hills and valleys in the potential that exist in the region of impurity atoms or defects. The localized states then act as the reservoir that maintains a constant Hall resistance for a finite range of magnetic field strengths.

It is rather unlikely that the quantized Hall effect will ever be employed in a device of great economic significance. The need for extremely strong magnetic fields and extremely low temperatures adds an expense that rules out most practical applications. Investigators will surely take advantage of the Hall plateaus in the laboratory, however, as precise standards of electrical resistance; in addition the quantized Hall effect will probably be incorporated into experiments establishing more precise values for the fundamental constants of quantum mechanics, electricity and magnetism. When they are combined with other measurements, these experiments may lead to more stringent tests of the fundamental theory of electromagnetic interactions, complementing tests of that theory carried out with high-energy accelerators.

The true importance of the quantized Hall effect does not lie in any of these applications, however, but rather in the new insight physicists have gained into the peculiar properties of systems of electrons in strong magnetic fields and into the hidden regularities implied by the mathematical laws of quantum mechanics. There is also value in the reminder that nature may well hold in store other surprising states of matter that none of us yet imagine.